

SMAPhysics.com

JEE Main Physics Practice Sheet — DPP

Topic: Centre of Mass of a Two-Particle System

Time Allowed: 60 Minutes — Max Marks: 80 — Marking Scheme: +4, -1

- Q1.** Two particles of masses $m_1 = 2$ kg and $m_2 = 3$ kg are located at positions $\vec{r}_1 = (2\hat{i} + 4\hat{j})$ m and $\vec{r}_2 = (7\hat{i} - \hat{j})$ m respectively. The position vector of their centre of mass is:
- (A) $(5\hat{i} + \hat{j})$ m
 (B) $(4\hat{i} + 2\hat{j})$ m
 (C) $(9\hat{i} + 3\hat{j})$ m
 (D) $(5\hat{i} + 2\hat{j})$ m
- Q2.** Two particles of masses m_1 and m_2 are separated by a fixed distance d . The distance of the centre of mass from the particle of mass m_1 is:
- (A) $\frac{m_1 d}{m_1 + m_2}$
 (B) $\frac{(m_1 + m_2)d}{m_1}$
 (C) $\frac{m_2 d}{m_1 + m_2}$
 (D) $\frac{m_2 d}{m_1}$
- Q3.** Two bodies of masses 2 kg and 4 kg are moving along the x -axis with velocities 6 m/s and -3 m/s respectively. The velocity of their centre of mass is:
- (A) 2 m/s
 (B) -1 m/s
 (C) 1 m/s
 (D) 0 m/s
- Q4.** Two particles of masses $m_1 = 3$ kg and $m_2 = 1$ kg are initially at rest at a separation of 12 m. They start moving towards each other under mutual gravitational attraction. The displacement of mass m_1 when the two particles collide is:
- (A) 3 m
 (B) 4 m
 (C) 9 m
 (D) 6 m
- Q5.** In a two-particle system with masses m_1 and m_2 , the sum of moments of the masses about their centre of mass ($\sum m_i \vec{r}_i^*$) is always:
- (A) Dependent on the external forces
 (B) Non-zero and positive
 (C) Dependent on the choice of origin
 (D) Strictly equal to zero
- Q6.** Two masses $m_1 = 4$ kg and $m_2 = 6$ kg are separated by a distance L . If the mass m_1 is displaced towards the right by a distance of 5 cm, by what distance and in which direction must mass m_2 be shifted so that the position of the centre of mass remains unchanged?
- (A) $\frac{10}{3}$ cm towards left
 (B) 5 cm towards left
 (C) $\frac{10}{3}$ cm towards right
 (D) $\frac{15}{2}$ cm towards right
- Q7.** Two masses m_1 and m_2 are connected by a light spring of spring constant k and placed on a smooth horizontal surface. Initially at rest, they are pulled apart and released. If at an instant the acceleration of m_1 is a_1 towards right, the acceleration of the centre of mass of the two-particle system is:
- (A) $\frac{m_1 a_1}{m_1 + m_2}$
 (B) 0
 (C) $\frac{m_2 a_1}{m_1}$
 (D) a_1
- Q8.** A system consists of two particles of masses 1 kg and 2 kg. The first particle has an acceleration $\vec{a}_1 = (2\hat{i} + 3\hat{j})$ m/s² and the second particle has an acceleration $\vec{a}_2 = (-\hat{i} + 6\hat{j})$ m/s². The magnitude of the acceleration of the centre of mass is:
- (A) 5 m/s²
 (B) $\sqrt{26}$ m/s²
 (C) 3 m/s²
 (D) 4 m/s²
- Q9.** If the internal forces acting between the two particles of an isolated system are non-zero, the motion of their centre of mass:
- (A) Must be accelerated
 (B) Follows a curved trajectory
 (C) Is completely unaffected by internal forces
 (D) Depends on the magnitude of internal forces
- Q10.** Two particles of masses $m_1 = 1$ kg and $m_2 = 3$ kg have position vectors $\vec{r}_1(t) = (2t^2\hat{i} + 3t\hat{j})$ m and $\vec{r}_2(t) = (t\hat{i} + 2t^2\hat{j})$ m respectively. The total external force acting on the two-particle system at any time t is:
- (A) $(2\hat{i} + 6\hat{j})$ N
 (B) $(4\hat{i} + 12\hat{j})$ N
 (C) $(4\hat{i} + 6\hat{j})$ N
 (D) $(8\hat{i} + 12\hat{j})$ N

- Q11.** In a diatomic molecule of carbon monoxide (CO), the distance between the carbon atom ($m_C = 12$ u) and the oxygen atom ($m_O = 16$ u) is $1.12 \text{ \AA}$. The distance of the centre of mass from the carbon atom is:
- (A) $0.48 \text{ \AA}$
 (B) $0.56 \text{ \AA}$
 (C) $0.64 \text{ \AA}$
 (D) $0.72 \text{ \AA}$
- Q12.** Two particles of masses $m_1 = 2$ kg and $m_2 = 3$ kg are projected vertically upward with speeds 30 m/s and 20 m/s respectively from the ground at $t = 0$. Neglecting air resistance, the maximum vertical height reached by the centre of mass of the two particles is ($g = 10 \text{ m/s}^2$):
- (A) 36.0 m
 (B) 28.8 m
 (C) 32.4 m
 (D) 25.0 m
- Q13.** A man of mass 50 kg stands at one end of a stationary wooden plank of mass 150 kg and length 4 m floating on calm water. If the man walks to the other end of the plank, the displacement of the plank relative to the water is:
- (A) 1.0 m
 (B) 1.33 m
 (C) 0.75 m
 (D) 2.0 m
- Q14.** The kinetic energy of a two-particle system of masses m_1 and m_2 with velocities $\vec{v}_1$ and $\vec{v}_2$ can be expressed as $K = \frac{1}{2}(m_1 + m_2)v_{\text{cm}}^2 + K_{\text{rel}}$. The internal kinetic energy K_{rel} relative to the centre of mass is:
- (A) $\frac{1}{2} \left(\frac{m_1 m_2}{m_1 + m_2} \right) |\vec{v}_1 + \vec{v}_2|^2$
 (B) $\frac{1}{2} (m_1 + m_2) |\vec{v}_1 - \vec{v}_2|^2$
 (C) $\frac{1}{2} \left(\frac{m_1 m_2}{m_1 + m_2} \right) |\vec{v}_1 - \vec{v}_2|^2$
 (D) $\frac{1}{2} \sqrt{m_1 m_2} |\vec{v}_1 - \vec{v}_2|^2$
- Q15.** Two particles of masses $m_1 = 2$ kg and $m_2 = 4$ kg are moving towards each other along the x -axis. In the centre of mass (C-frame) reference frame, the ratio of the magnitude of linear momentum of m_1 to that of m_2 ($p_{1,\text{cm}} : p_{2,\text{cm}}$) is:
- (A) $1 : 2$
 (B) $2 : 1$
 (C) $1 : 1$
 (D) $1 : 4$
- Q16.** Two particles of masses 2 kg and 3 kg are located at coordinates $(1, 2)$ and $(6, 7)$ respectively. If the origin of the coordinate system is shifted to $(2, 3)$, the coordinates of their centre of mass in the new reference frame are:
- (A) $(2, 2)$
 (B) $(4, 5)$
 (C) $(3, 3)$
 (D) $(1, 2)$
- Q17.** Two particles of equal mass m are projected simultaneously from the ground with the same initial speed u at angles 30° and 60° with the horizontal in the same vertical plane. The trajectory traced by their centre of mass before either particle lands is:
- (A) A vertical straight line
 (B) A parabola
 (C) A circular arc
 (D) An inclined straight line
- Q18.** Two particles of masses m_1 and m_2 have initial velocities $\vec{u}_1$ and $\vec{u}_2$. A constant external force $\vec{F}$ acts on m_1 while no external force acts on m_2 . The velocity of the centre of mass after time t is:
- (A) $\frac{m_1 \vec{u}_1 + m_2 \vec{u}_2}{m_1 + m_2} + \frac{\vec{F}t}{m_1}$
 (B) $\frac{m_1 \vec{u}_1 + m_2 \vec{u}_2 + \vec{F}t}{m_1 + m_2}$
 (C) $\frac{\vec{u}_1 + \vec{u}_2}{2} + \frac{\vec{F}t}{m_1 + m_2}$
 (D) $\frac{m_1 \vec{u}_1 + m_2 \vec{u}_2}{m_1 + m_2} + \frac{\vec{F}t}{m_2}$
- Q19.** A mass $m_1 = 5$ kg is located at $x = 0$ and another mass $m_2 = 15$ kg is located at $x = 20$ m. If mass m_1 is moved by 4 m along $+x$, the distance by which the centre of mass shifts is:
- (A) 1.0 m
 (B) 2.0 m
 (C) 3.0 m
 (D) 0.5 m
- Q20.** Two particles of masses m and $2m$ are connected by a massless rigid rod of length L . The moment of inertia of this two-particle system about an axis passing through its centre of mass and perpendicular to the rod is:
- (A) $\frac{1}{3}mL^2$
 (B) $\frac{3}{2}mL^2$
 (C) $\frac{2}{3}mL^2$
 (D) $\frac{2}{9}mL^2$

SMAPhysics.com — Answer Key & Detailed Solutions

Centre of Mass of a Two-Particle System

Q.No.	Ans	Q.No.	Ans	Q.No.	Ans	Q.No.	Ans
1	A	6	A	11	C	16	A
2	C	7	B	12	B	17	B
3	D	8	A	13	A	18	B
4	A	9	C	14	C	19	A
5	D	10	B	15	C	20	C

Detailed Step-by-Step Solutions

Sol 1. Option (A)

Position vector of the centre of mass:

$$\vec{r}_{\text{cm}} = \frac{m_1 \vec{r}_1 + m_2 \vec{r}_2}{m_1 + m_2}$$

Substituting values:

$$\vec{r}_{\text{cm}} = \frac{2(2\hat{i} + 4\hat{j}) + 3(7\hat{i} - \hat{j})}{2 + 3} = \frac{(4\hat{i} + 8\hat{j}) + (21\hat{i} - 3\hat{j})}{5}$$

$$\vec{r}_{\text{cm}} = \frac{25\hat{i} + 5\hat{j}}{5} = (5\hat{i} + \hat{j}) \text{ m}$$

Sol 2. Option (C)

Taking m_1 at the origin ($x_1 = 0$) and m_2 at $x_2 = d$:

$$x_{\text{cm}} = \frac{m_1(0) + m_2(d)}{m_1 + m_2} = \frac{m_2 d}{m_1 + m_2}$$

Sol 3. Option (D)

Velocity of the centre of mass:

$$v_{\text{cm}} = \frac{m_1 v_1 + m_2 v_2}{m_1 + m_2} = \frac{2(6) + 4(-3)}{2 + 4} = \frac{12 - 12}{6} = 0 \text{ m/s}$$

Sol 4. Option (A)

Since no external horizontal force acts on the system, the centre of mass remains stationary ($\Delta x_{\text{cm}} = 0$).

Taking m_1 at $x = 0$ and m_2 at $x = 12$ m:

$$x_{\text{cm}} = \frac{3(0) + 1(12)}{3 + 1} = \frac{12}{4} = 3 \text{ m}$$

Because both particles collide at the centre of mass position, the displacement of m_1 is:

$$\Delta x_1 = 3 \text{ m} - 0 \text{ m} = 3 \text{ m}$$

Sol 5. Option (D)

By definition of the centre of mass, the position vector of particle i relative to the centre of mass is $\vec{r}'_i = \vec{r}_i - \vec{r}_{\text{cm}}$.

Sum of mass moments:

$$\begin{aligned} \sum m_i \vec{r}'_i &= \sum m_i (\vec{r}_i - \vec{r}_{\text{cm}}) \\ &= \sum m_i \vec{r}_i - \vec{r}_{\text{cm}} \sum m_i \\ &= M \vec{r}_{\text{cm}} - M \vec{r}_{\text{cm}} = \vec{0} \end{aligned}$$

Hence, it is identically zero for all systems.

Sol 6. Option (A)

For the position of centre of mass to remain unchanged:

$$\Delta x_{\text{cm}} = \frac{m_1 \Delta x_1 + m_2 \Delta x_2}{m_1 + m_2} = 0 \implies m_1 \Delta x_1 + m_2 \Delta x_2 = 0$$

Given $m_1 = 4$ kg, $\Delta x_1 = +5$ cm, $m_2 = 6$ kg:

$$4(+5) + 6(\Delta x_2) = 0 \implies \Delta x_2 = -\frac{20}{6} = -\frac{10}{3} \text{ cm}$$

The negative sign indicates mass m_2 must be shifted towards the left by $\frac{10}{3}$ cm.

Sol 7. Option (B)

The spring force between m_1 and m_2 is an internal action-reaction force. Since the horizontal surface is smooth, the net external force on the system is zero ($\sum \vec{F}_{\text{ext}} = \vec{0}$).

Therefore:

$$\vec{a}_{\text{cm}} = \frac{\sum \vec{F}_{\text{ext}}}{m_1 + m_2} = 0$$

Sol 8. Option (A)

Acceleration of the centre of mass:

$$\vec{a}_{\text{cm}} = \frac{m_1 \vec{a}_1 + m_2 \vec{a}_2}{m_1 + m_2} = \frac{1(2\hat{i} + 3\hat{j}) + 2(-\hat{i} + 6\hat{j})}{1 + 2}$$

$$\vec{a}_{\text{cm}} = \frac{(2\hat{i} + 3\hat{j}) + (-2\hat{i} + 12\hat{j})}{3} = \frac{0\hat{i} + 15\hat{j}}{3} = 5\hat{j} \text{ m/s}^2$$

Magnitude $|\vec{a}_{\text{cm}}| = 5 \text{ m/s}^2$.

Sol 9. Option (C)

According to Newton's third law, internal forces between particles in a system occur in equal and opposite pairs ($\vec{F}_{12} = -\vec{F}_{21}$). Their vector sum is zero, so internal forces have zero net contribution to $\vec{a}_{\text{cm}}$. Thus, the motion of the centre of mass is completely independent of internal forces.

Sol 10. Option (B)

Accelerations of the individual particles are:

$$\vec{a}_1 = \frac{d^2 \vec{r}_1}{dt^2} = \frac{d^2}{dt^2} (2t^2 \hat{i} + 3t \hat{j}) = 4\hat{i} \text{ m/s}^2$$

$$\vec{a}_2 = \frac{d^2 \vec{r}_2}{dt^2} = \frac{d^2}{dt^2} (t \hat{i} + 2t^2 \hat{j}) = 4\hat{j} \text{ m/s}^2$$

The net external force on the system is:

$$\vec{F}_{\text{ext}} = m_1 \vec{a}_1 + m_2 \vec{a}_2 = 1(4\hat{i}) + 3(4\hat{j}) = (4\hat{i} + 12\hat{j}) \text{ N}$$

Sol 11. Option (C)

Taking the carbon atom at the origin ($x_{\text{C}} = 0$) and oxygen atom at $x_{\text{O}} = 1.12$ Å:

$$x_{\text{cm}} = \frac{m_{\text{C}}(0) + m_{\text{O}}(1.12)}{m_{\text{C}} + m_{\text{O}}} = \frac{16 \times 1.12}{28} = 0.64 \text{ Å}$$

Sol 12. Option (B)

Initial vertical velocity of the centre of mass:

$$u_{\text{cm}} = \frac{2(30) + 3(20)}{2 + 3} = \frac{120}{5} = 24 \text{ m/s}$$

Since both particles experience uniform downward acceleration due to gravity $g = 10 \text{ m/s}^2$, the centre of mass has constant downward acceleration $a_{\text{cm}} = -g$.

Maximum height attained by the centre of mass:

$$H_{\text{max}} = \frac{u_{\text{cm}}^2}{2g} = \frac{24^2}{2 \times 10} = \frac{576}{20} = 28.8 \text{ m}$$

Sol 13. Option (A)

Let the displacement of the plank relative to water be x in the opposite direction. Since no external horizontal force acts:

$$m_{\text{man}}(L - x) - m_{\text{plank}}(x) = 0 \implies x = \frac{m_{\text{man}}L}{m_{\text{man}} + m_{\text{plank}}}$$

Substituting given values:

$$x = \frac{50 \times 4}{50 + 150} = \frac{200}{200} = 1.0 \text{ m}$$

Sol 14. Option (C)

By König's theorem for a two-body system, total kinetic energy is decomposed into the kinetic energy of the centre of mass and relative motion:

$$K = \frac{1}{2}Mv_{\text{cm}}^2 + \frac{1}{2}\mu v_{\text{rel}}^2$$

where $\mu = \frac{m_1 m_2}{m_1 + m_2}$ is the reduced mass and $\vec{v}_{\text{rel}} = \vec{v}_1 - \vec{v}_2$.

Thus, $K_{\text{rel}} = \frac{1}{2} \left(\frac{m_1 m_2}{m_1 + m_2} \right) |\vec{v}_1 - \vec{v}_2|^2$.

Sol 15. Option (C)

In the centre of mass reference frame, total momentum is identically zero by definition:

$$\vec{P}_{\text{cm}} = \vec{p}_{1,\text{cm}} + \vec{p}_{2,\text{cm}} = \vec{0} \implies \vec{p}_{1,\text{cm}} = -\vec{p}_{2,\text{cm}}$$

Therefore, the magnitudes are always strictly equal:

$$|\vec{p}_{1,\text{cm}}| = |\vec{p}_{2,\text{cm}}| \implies \text{Ratio} = 1 : 1$$

Sol 16. Option (A)

In the original frame:

$$x_{\text{cm}} = \frac{2(1) + 3(6)}{2 + 3} = \frac{20}{5} = 4$$

$$y_{\text{cm}} = \frac{2(2) + 3(7)}{2 + 3} = \frac{25}{5} = 5$$

When the origin is shifted to $(x_0, y_0) = (2, 3)$:

$$x'_{\text{cm}} = x_{\text{cm}} - x_0 = 4 - 2 = 2$$

$$y'_{\text{cm}} = y_{\text{cm}} - y_0 = 5 - 3 = 2$$

So the coordinates in the new frame are $(2, 2)$.

Sol 17. Option (B)

Both particles move under gravity with identical downward acceleration $\vec{g} = -g\hat{j}$.

Thus, the acceleration of the centre of mass is $\vec{a}_{\text{cm}} = \vec{g}$.

The initial velocity of COM is:

$$\vec{u}_{\text{cm}} = \frac{u(\cos 30^\circ \hat{i} + \sin 30^\circ \hat{j}) + u(\cos 60^\circ \hat{i} + \sin 60^\circ \hat{j})}{2}$$

Since $\vec{u}_{\text{cm}}$ has non-zero horizontal and vertical components while experiencing constant vertical downward acceleration $\vec{g}$, its trajectory is a parabola.

Sol 18. Option (B)

Initial velocity of COM: $\vec{v}_{\text{cm}}(0) = \frac{m_1 \vec{u}_1 + m_2 \vec{u}_2}{m_1 + m_2}$.

Acceleration of COM: $\vec{a}_{\text{cm}} = \frac{\sum \vec{F}_{\text{ext}}}{m_1 + m_2} = \frac{\vec{F}}{m_1 + m_2}$.

Using $\vec{v}_{\text{cm}}(t) = \vec{v}_{\text{cm}}(0) + \vec{a}_{\text{cm}}t$:

$$\vec{v}_{\text{cm}}(t) = \frac{m_1 \vec{u}_1 + m_2 \vec{u}_2 + \vec{F}t}{m_1 + m_2}$$

Sol 19. Option (A)

Shift in the position of the centre of mass:

$$\Delta x_{\text{cm}} = \frac{m_1 \Delta x_1 + m_2 \Delta x_2}{m_1 + m_2} = \frac{5(4) + 15(0)}{5 + 15} = \frac{20}{20} = 1.0 \text{ m}$$

Sol 20. Option (C)

Distance of m from COM: $r_1 = \frac{2m}{m+2m}L = \frac{2}{3}L$.

Distance of $2m$ from COM: $r_2 = \frac{m}{m+2m}L = \frac{1}{3}L$.

Moment of inertia about the COM axis:

$$\begin{aligned} I &= m \left(\frac{2}{3}L \right)^2 + 2m \left(\frac{1}{3}L \right)^2 \\ &= \frac{4}{9}mL^2 + \frac{2}{9}mL^2 = \frac{2}{3}mL^2 \end{aligned}$$